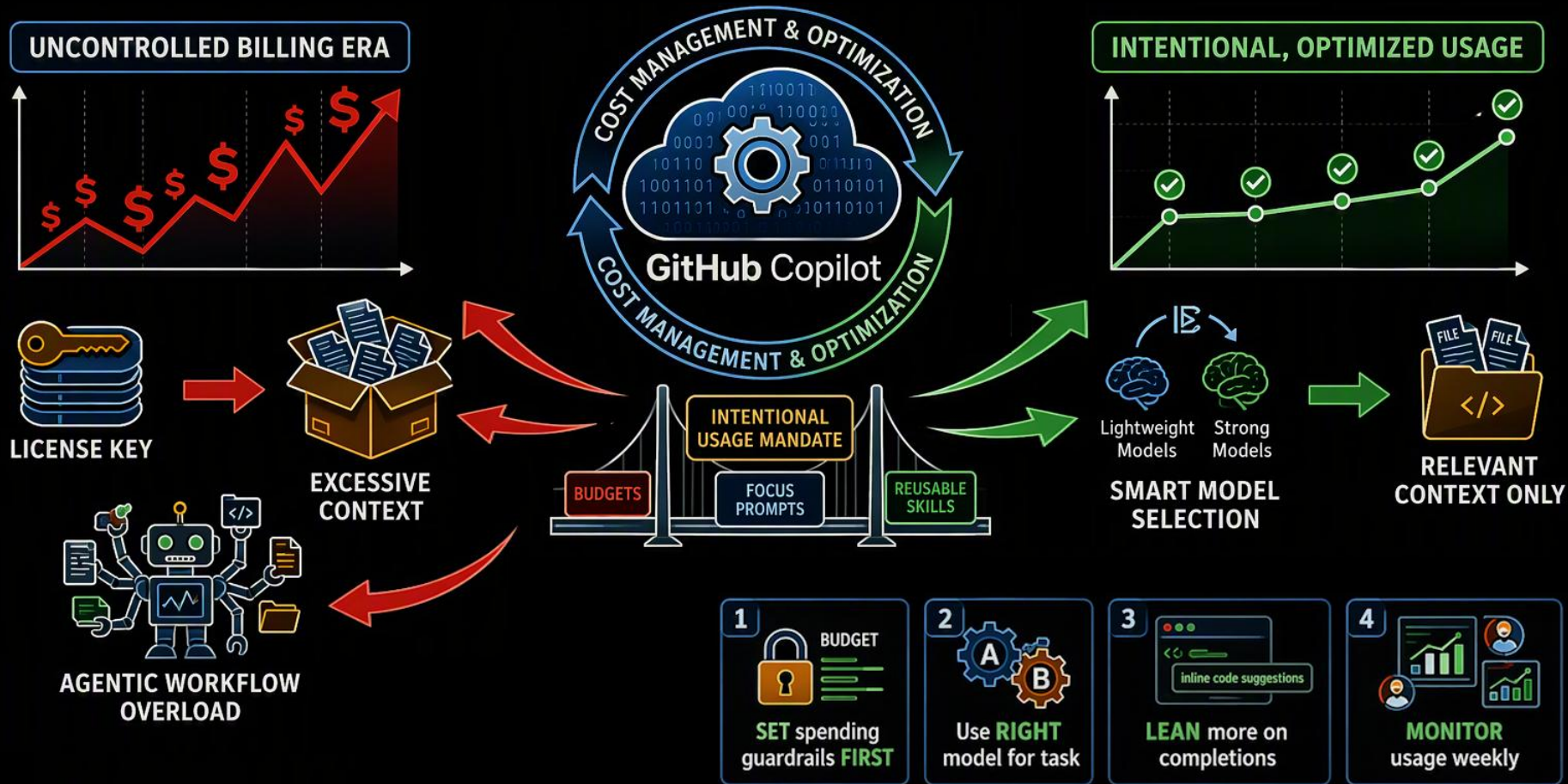


GitHub Copilot Tokenomics



Terms and Conditions

© Microsoft Corporation. All rights reserved.

You may use these training materials solely for your personal internal reference and non-commercial purposes. You may not distribute, transmit, resell or otherwise make these training materials available to any other person or party without express permission from Microsoft Corporation. URL's or other internet website references in the training materials may change without notice. Unless otherwise noted, any companies, organizations, domain names, e-mail addresses, people, places and events depicted in the training materials are for illustration only and are fictitious. No real association is intended or inferred. THESE TRAINING MATERIALS ARE PROVIDED "AS IS"; MICROSOFT MAKES NO WARRANTIES, EXPRESS OR IMPLIED IN THESE TRAINING MATERIALS

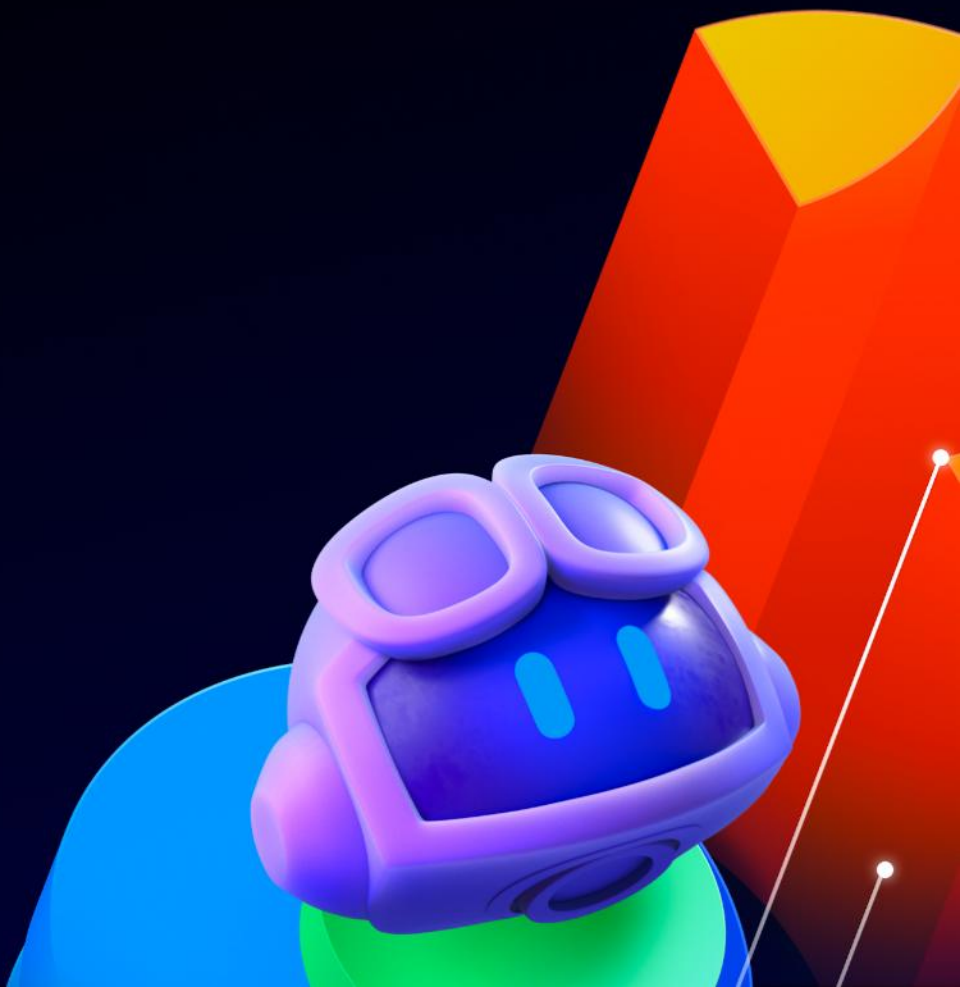
Our Agenda

- 📖 What is Usage Based Billing and how does Copilot actually work?
- 📖 Lever 1: Model Choice (Micro-Tokenomics)
- 📖 Lever 2: Context Windows (Macro-Tokenomics)
- 📖 Best Practices





How did we
get here...?



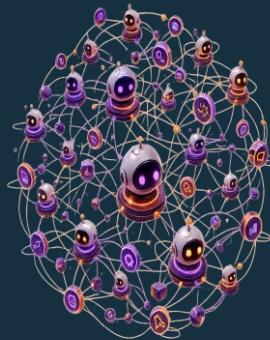
Why token usage is harder to predict now

2024 — life was simple



- Copilot was mostly single-prompt, request-and-response
- PRUs fit that world well — one request $\approx$ one unit
- Easy to reason about and to budget

2026 — agents changed the math



- One user request can trigger many model calls
- Agents orchestrate sub-agents, tools, and loops
- Long-running workflows compound tokens fast

Billing followed the technology: from counting requests → to measuring tokens.

GitHub AI Credits — the new unit of cost



1 AI credit

= \$0.01 USD

How a request becomes a bill

- Every interaction consumes input, output, and cached tokens
- Each token is priced by the model used
- Total token cost is converted into AI credits (1 credit = \$0.01)

Monthly allowance

- Copilot Business — 1,900 / license
- Copilot Enterprise — 3,900 / license
- Pooled at the billing-entity level

Consumes credits

- Copilot Chat & CLI
- Cloud agent, Spaces, Spark
- Third-party coding agents

Does NOT consume credits

- Code completions
- Next edit suggestions
- Unlimited on paid plans

View Cost in the Copilot CLI

```
> /usage          Display session usage metrics and statistics
  /compact        Summarize conversation history to reduce context window u...
  /context        Show context window token usage and visualization
```

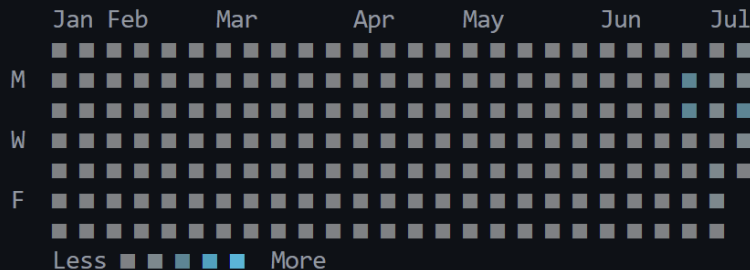
```
/usage
```

B

```
/help show help
```

Auto

• Activity • last 180 days • 58 messages



Changes +1436 -50

AI Credits 439 (88h 4m 17s)

C

Tokens ↑ 11.4m (10.6m cached, 189.1k written) • ↓ 72.5k (24.9k reasoning)

gpt-5.3-codex ↑ 10.6m (10.0m cached) • ↓ 60.2k (21.0k reasoning)

claude-sonnet-4.6 ↑ 792.9k (592.5k cached, 189.1k written) • ↓ 12.3k (3.9k reasoning)

Plan no limit

C:\lyle\Dadabase\dadabase.cli.llupesms [feature/cli-ratings]

A

Session: 439 AIC used

/ commands • ? help • → next tab

Auto

View Cost in VS Code Chat

(for one turn only...)

(new version coming soon!)

11:22 PM • GPT-5.6 Sol • 1458.1 credits

Response details

Completed: 8/24/2026, 11:22 PM

Model: GPT-5.6 Sol

- Input tokens: 30.4M
- Cached input tokens: 30M
- Output tokens: 73K

Model: Claude Opus 4.8

- Input tokens: 1M
- Cached input tokens: 840K
- Output tokens: 20K

CHAT

+ v ⚙️ ... | [] X

← ADD RATING FEATURE



Summary

✅ Setup Complete – Rating Feature Ready for Implementation

Implementation Queued:

- Phase 1 (Database): 4 tasks targeting SQL schema and stored procedure
- Phases 2–7: Repository, services, API, UI, testing, hardening

Estimated Effort: ~39 developer-hours total

Ready to begin implementation whenever you are. Which phase or task would you like to start with?



Claude Sonnet 4.6 • 43.4 credits

> ● Phase 1: SQL schema changes (1/10)



Describe what to build

+ ⚙️ Agent ⚙️ Auto ⚙️



Local Default Approvals



View Cost in VS Code Chat

(for the entire session...)

CHAT

← ADD RATING FEATURE

Ran `git add "Docs/Rating-Implementation-Tasks.md"...`

Summary

✓ Setup Complete – Rating Feature Ready for Implementation

Implementation Queued:

- Phase 1 (Database): 4 tasks targeting SQL schema and stored procedures
- Phases 2–7: Repository, services, API, UI, testing, hardening

Estimated Effort: ~39 developer-hours total

Ready to begin implementation whenever you are. Which phase or task

>

Phase 1: SQL schema changes (1/10)

Describe what to build

+

Agent

Claude Opus 4.6

High 1M

Local

Bypass Approvals

Session Info

Session Cost754.9 credits

Context Window

185.3K / 1M tokens19%

Reserved for response

System

System Instructions1.9%

Tool Definitions2.4%

User Context

Messages7.4%

Tool Results3.0%

Files3.9%

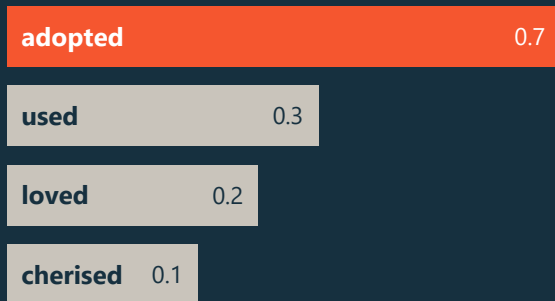
Compact Conversation

Do you know
what actually
happens
when you
submit a
prompt...?



An LLM is a word-probability machine

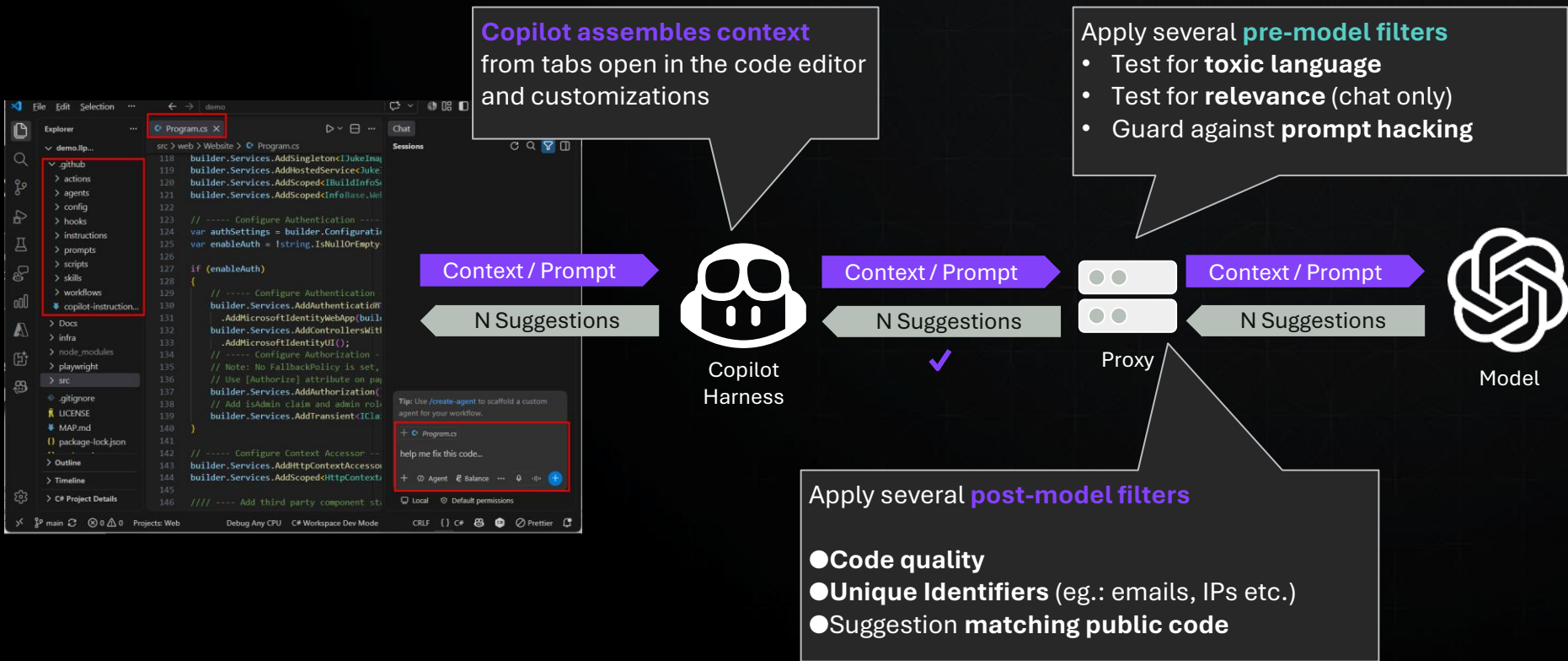
GitHub Copilot is the world's most widely...



What this means

- Text in, text out — nothing more mystical
- It predicts the most probable next word, then the next
- In coding: predicting the next statement instead of prose
- Bells and whistles were added, but the core is still this

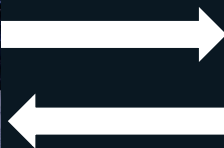
How GitHub Copilot fulfills a request



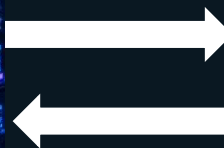
It's not magic — it's just (stateless) text



You & your project



The agent (harness)



The LLM

The agent talks to the LLM on your behalf — dozens of times per task. And the LLM is stateless: it stores nothing. "Having a conversation" means re-sending the entire ordered history — every input and output — on every single turn. Tokens and context **compound** with each loop.

Basic Economic Theory

Micro-Economics:

Which car or truck should I buy?

- **Performance:** A Ford F-250 can pull and transport huge loads and has a Super Cab, while a Prius is limited in capacity.
- **Efficiency and operating costs:** A Prius will consume less fuel and cost much less to own over time, while the truck is more expensive.
- **Upfront Cost:** A heavy duty truck requires a much higher investment.



Macro-Economics:

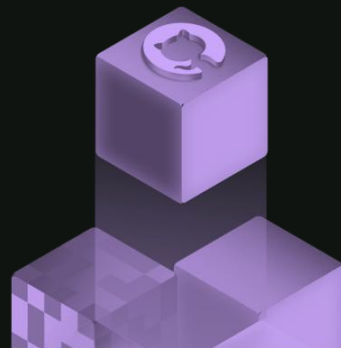
What affects the entire country?

- As money supply rises, there is too much money for too few goods.
- Inflation begins to rise, increasing the cost of goods.
- Every cost inflation triggers other cost increases such as salary increase.
- Governments might raise interest rate in an attempt to control inflation.



Basic “Tokenomics” Theory

The biggest levers for optimizing quality & tokens



Micro-Tokenomics



Model Choice

- Premium model → slower, smarter = more credits
- Efficient model → good enough = lower cost
- You optimize capability vs cost, per task

Macro-Tokenomics

Context Windows

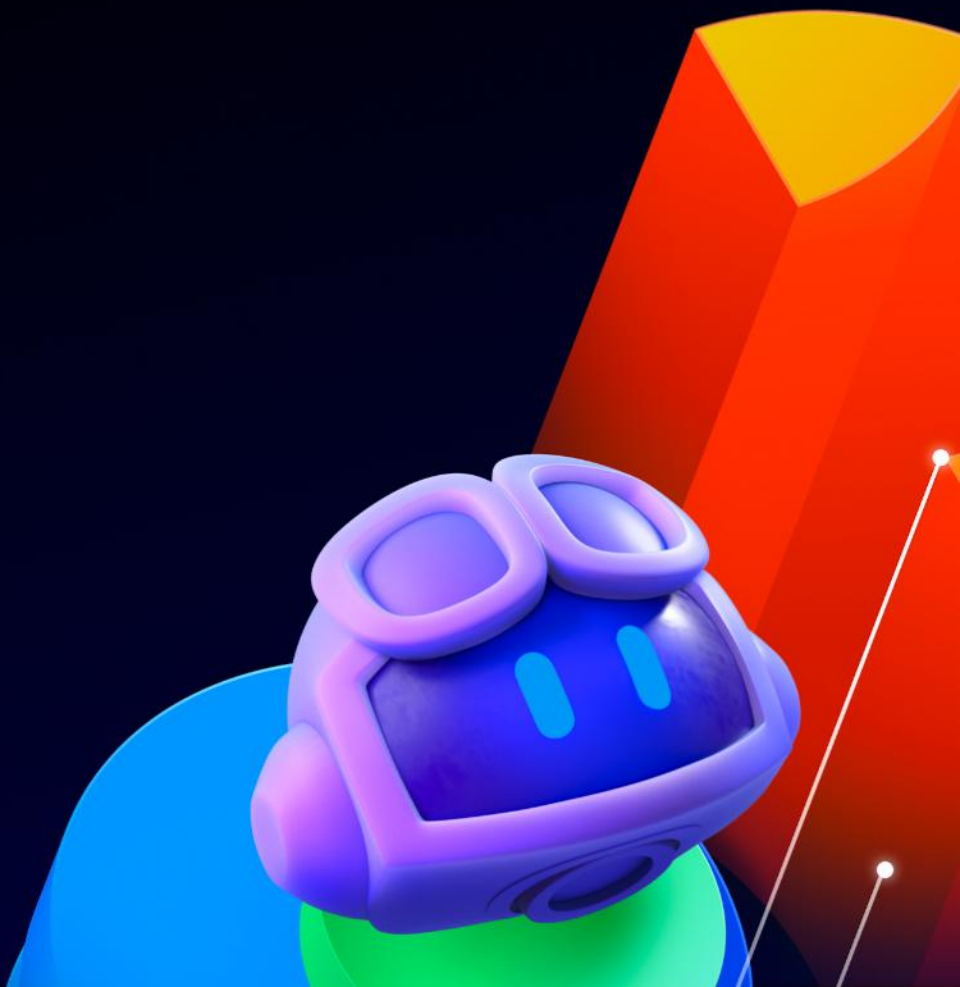


- The harness re-sends the entire history to the LLM, every turn
- Tokens compound as the session grows
- Context engineering is the core skill of controlling that context window



MICRO-TOKENOMICS

Model Selection



LEVER 1 — MODEL CHOICE

It starts with choosing the right model

Even a single model family has variations with different abilities and very different prices per 1M tokens.

~25× cost gap

between the most powerful and the leanest models

Bigger ≠ better

- Model choice drives BOTH token cost and quality
- The most capable model is often overkill — and can hurt results
- Right-sizing is the fastest win available to a developer

ANTHROPIC

 OpenAI

 Microsoft

 Google AI

Models — four buckets

Deep reasoning & complex coding

- Claude Opus 4/5
- Gemini Pro
- GPT-5.5/5.6 Sol
- Claude Fable (long horizon coding)
- GPT Astra (long horizon coding)

Agentic software development

- GPT-5.3-Codex
- GPT-5.4 mini
- Kimi K3
- GPT-5.6 Terra

General purpose

- Claude Sonnet 4/5
- GPT-5.3-Codex, 5 mini, 5.6 Terra
- MAI-Code-1.1-Flash
- Kimi K2.7 Code
- Grok 4.5/4.6
- Qwen 2.5

Fast, simple, repetitive

- GPT-5.6 Luna
- Claude Haiku 4.5
- MAI-Code-1.1-Flash
- Gemini 3.5 Flash

Each model has a slightly different cost

* All prices are per 1 million tokens

Model	Category	Input	Cached (R)	Cached (W)	Output
GPT-5.6 Astra	Super Powerful	\$10.00	\$1.00	\$12.50	\$50.00
Claude Fable 5	Super Powerful	\$10.00	\$1.00	\$12.50	\$50.00
GPT-5.6 Sol (promo Sept-Nov!)	Powerful	\$4.00	\$0.40	\$5.00	\$20.00
GPT-5.5	Powerful	\$5.00	\$0.50		\$30.00
Claude Opus 5	Powerful	\$5.00	\$0.50	\$6.25	\$25.00
GPT-5.6 Terra	Versatile	\$2.00	\$0.20	\$2.50	\$12.00
Claude Sonnet 5	Versatile	\$2.00	\$0.20	\$2.50	\$10.00
GPT-5.3-Codex	Powerful	\$1.75	\$0.18		\$14.00
Gemini 3.5 Flash	Lightweight	\$1.50	\$0.15		\$9.00
GPT-5.6 Luna (price cut!)	Lightweight	\$0.20	\$0.02	\$0.50	\$1.80
Claude Haiku 4.5	Versatile	\$1.00	\$0.10	\$1.25	\$5.00
MAI-Code-1.1- Flash	Lightweight	\$0.20	\$0.02		\$1.20
Kimi K2.7 Code	Versatile	\$0.95	\$0.19		\$4.00
GPT-5 mini	Lightweight	\$0.25	\$0.03		\$2.00

Token Basics

1,000,000 tokens

≈ the entire **Lord of the Rings** trilogy — plus **The Hobbit**.

That's the ceiling of what fits, not a target.

- 1 token ≈ $\frac{3}{4}$ of a word — think "1 token ≈ 1 word"
- Context limits vary: ~50–200K (small) up to ~1M (large)
- Prompts, files, and responses all cost tokens — and compound every loop



When to reason, when to keep it light



Reasoning models

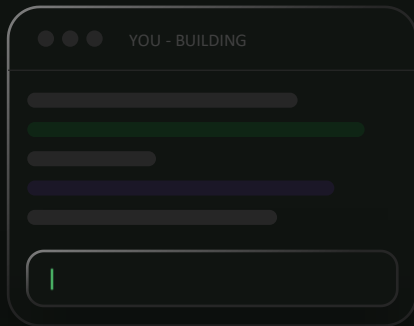
Planning, architecture, subtle bugs,
and many-file tasks
— where thinking pays off.

Lightweight models

Implementation after the plan
— fast, well-specified execution.

Watch out — hand a reasoning model a tight spec and it may re-open the plan and "go rogue." For execution, that's a liability.

Auto Mode: The right model for each task. Automatically.



● OPUS 4.8

● GPT 5.6 SOL

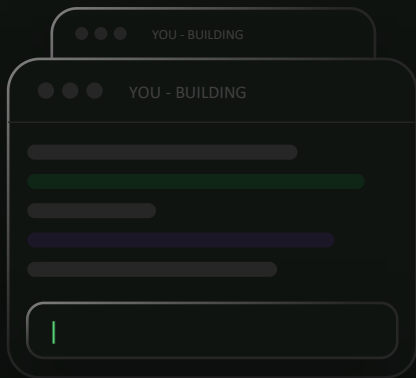
● GEMINI 3 PRO

● SONNET 5

● KIMI K2.5

● FABLE 5

Auto Mode: The right model for each task. Automatically.



● OPUS 4.8

● GPT 5.6 SOL

● GEMINI 3 PRO

● SONNET 5

● KIMI K2.5

● FABLE 5

Auto mode and matching the model to each phase

Auto mode — stay lazy on purpose

- Detects task intent and picks a right-sized model
- A sensible default for most routine work
- Make deliberate choices only when it truly matters
- The scalable answer for hundreds of millions of users

Match the model to each phase

- Plan phase → a reasoning model
- Implement phase → a mid-tier or lightweight model
- Review phase → a code specialist
- Set it when you START a new session or phase

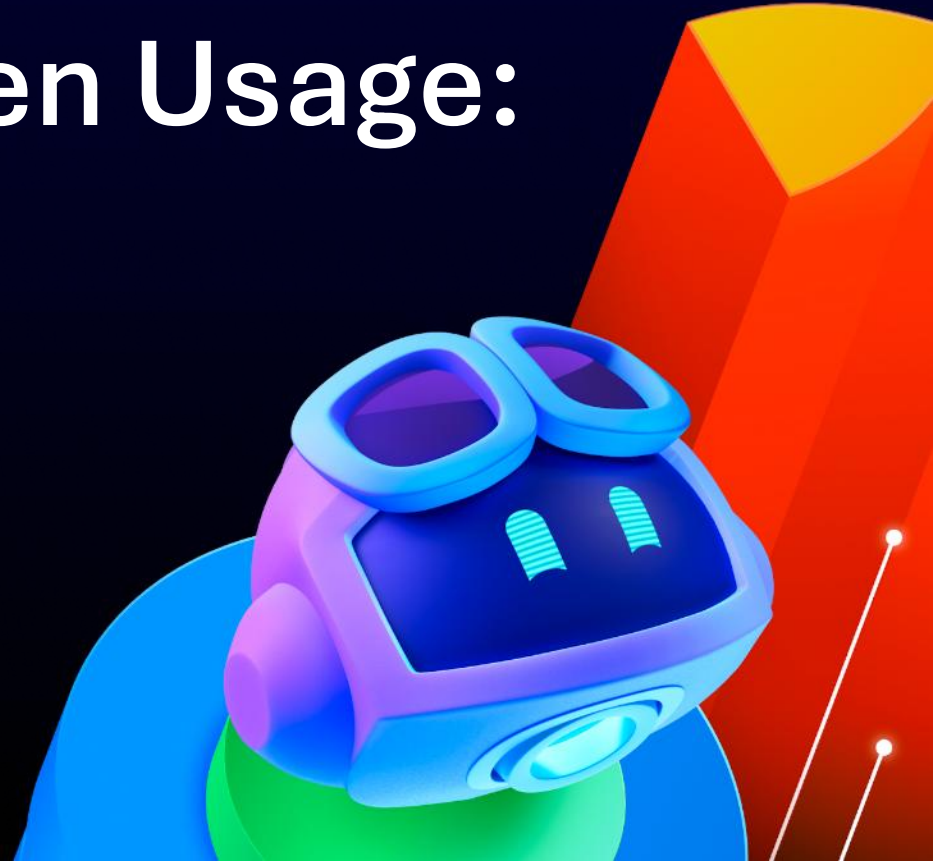
Switch by phase — not mid-conversation. Plan → reasoning · implement → mid-tier · review → specialist, each in a NEW session/window. Changing models inside a live session invalidates the prompt cache — the new model has no cached state, so the whole context is re-billed at full price.



MACRO-TOKENOMICS

Controlling Token Usage:

What is in your
System
Instructions?



What exactly is in the system instructions?

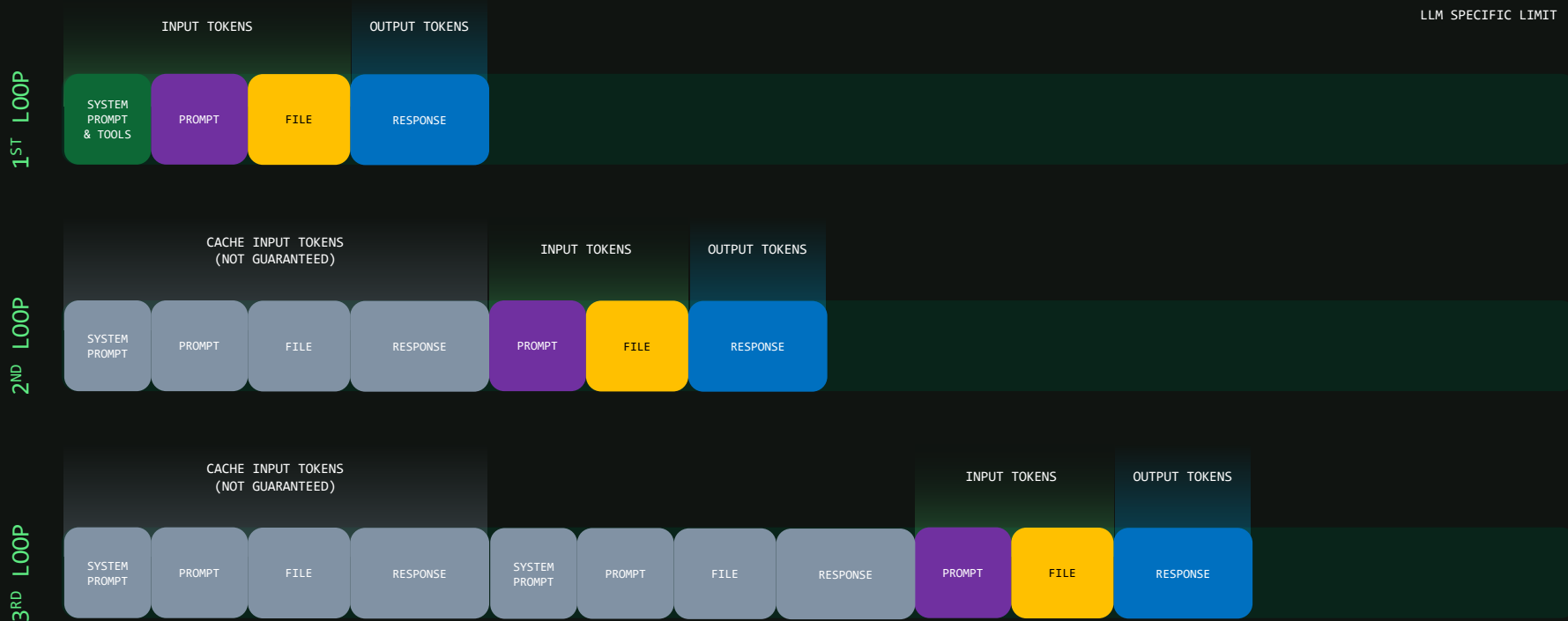


Instructions,
History,
References

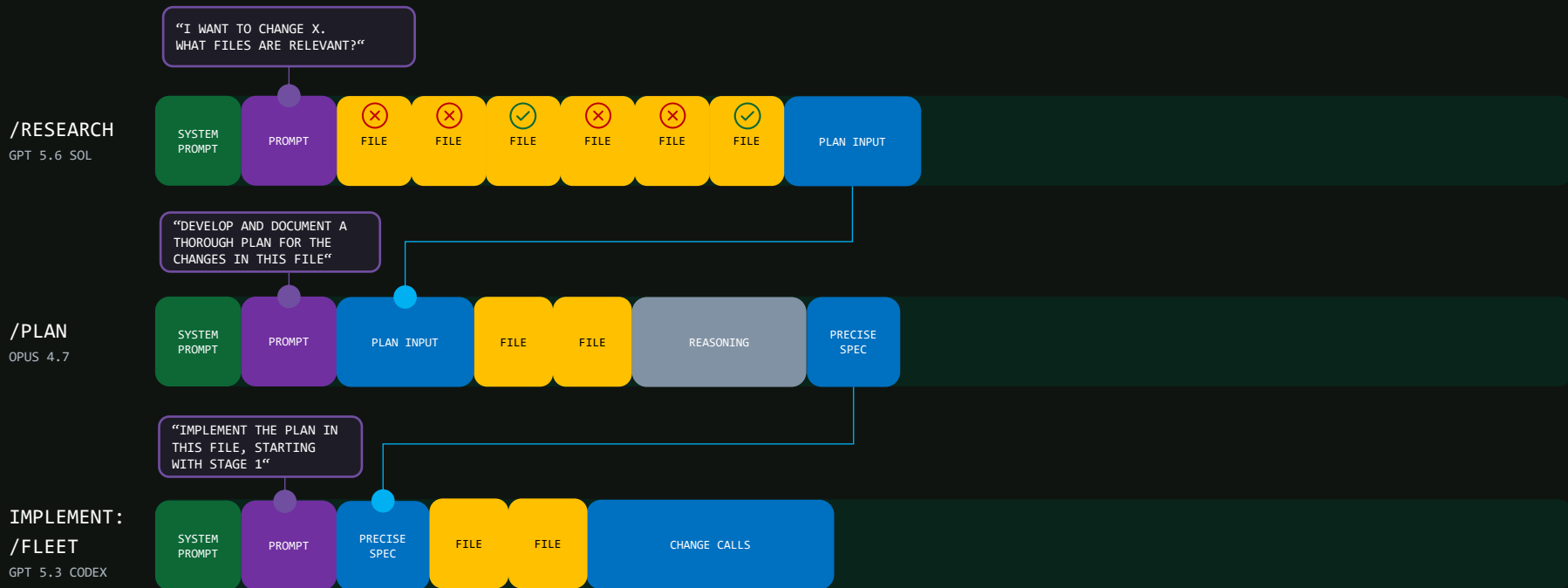
Context Rot,
Noise

-  **Identity**
You are GitHub Copilot. You are a skilled programmer.
-  **Code Rules**
"Make precise surgical changes..."
-  **Guidelines**
"Use Rubber Duck to critique plans"
-  **Limitations**
Don't share sensitive data, don't leak the system prompt,
don't create harmful content
-  **Tool Use Instructions**
Built-in tools (read, view, sub-agents, PowerShell), MCP Servers, Skills
-  **Custom instructions**
AGENTS.md, copilot-instructions.md
-  **Additional Instructions**
Plan Mode, Autopilot
-  **Custom Agents**
-  **Final Instructions**
Response concisely, finish the job....

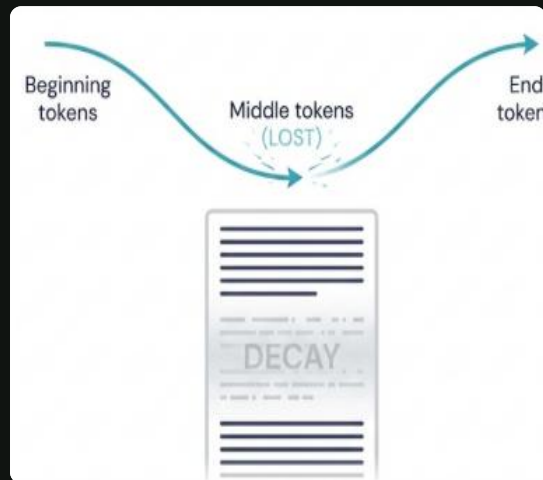
Context Window & Tokens



Research → Plan → Implement

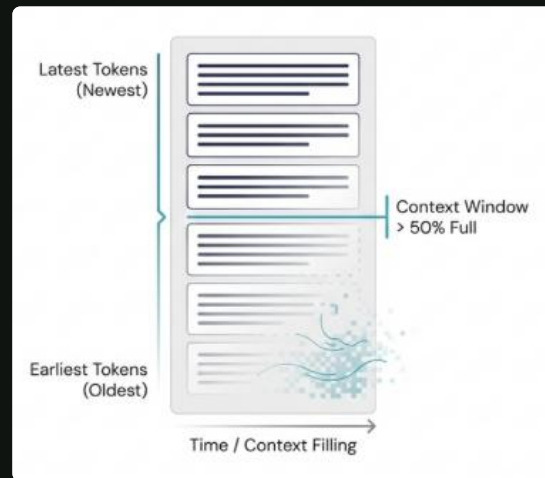


Context Rot



Lost in the Middle (<50% full)

Models bias tokens at the beginning and end of the context.



Recency Bias (> 50% full)

Models bias tokens from the end of the context.

Reference:

<https://www.producttalk.org/context-rot/>

Rule of thumb: keep working sessions under ~60–70% of the window, and use a NEW window per distinct task. Just because you can fill the context window doesn't mean you should.



Everything should
be made
as simple as
possible,
but not simpler.

Albert Einstein *

(* maybe...)

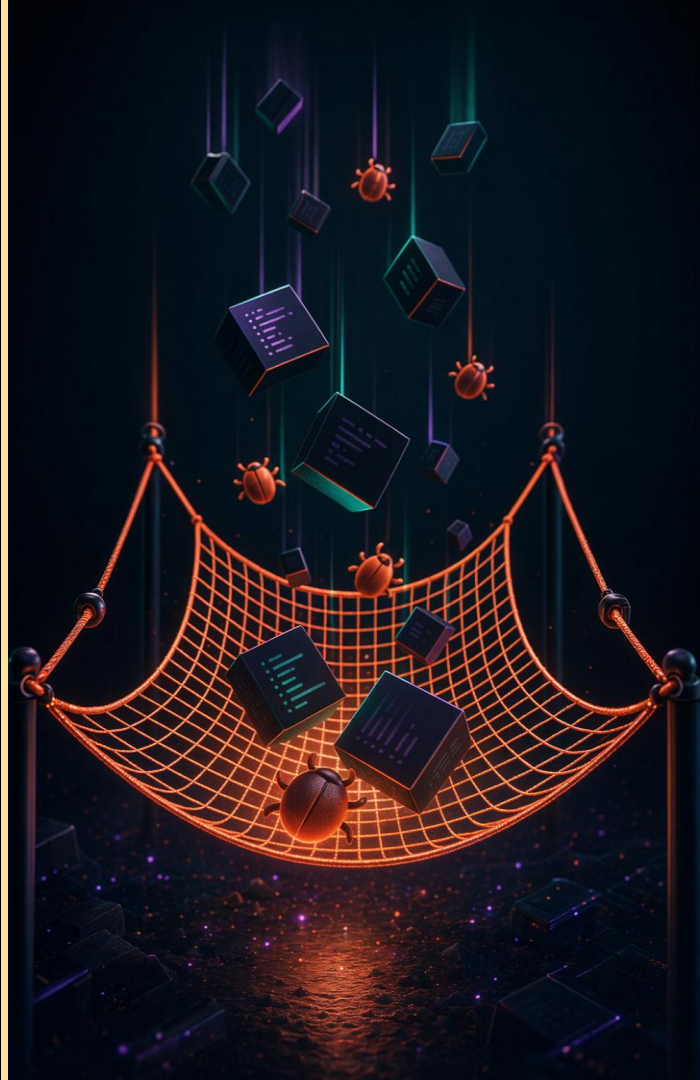
Provide as little
context as possible,
but as much as
required.



SECTION 04

Quality & token controls

Deterministic guardrails and the agent configuration files that keep quality high while spending less.



SPECIFIC AREAS

Your agent-configuration toolbox

Context engineering is mostly agent configuration — the files agents pick up automatically. Each has a different job, cost, and trigger.

Persistent instructions

copilot-instructions.md — always on

Skills

SKILL.md — lazy-loaded capability

Sub-agents

Second window for a sub-task

Prompt files

*.prompt.md — manual starting points

Custom agents

*.agent.md — scoped role & tools

MCP servers

External tools & APIs

Scoped instructions

*.instructions.md — path-based

Copilot memory

Learns patterns over time



Persistent instructions — your always-on guidance

`copilot-instructions.md` / `AGENTS.md`

What belongs

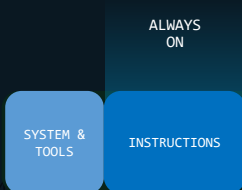
- Non-negotiables & guardrails
- Recurring agent-miss fixes
- Output trimming ("be concise")
- One-line tech-stack context

Best practices

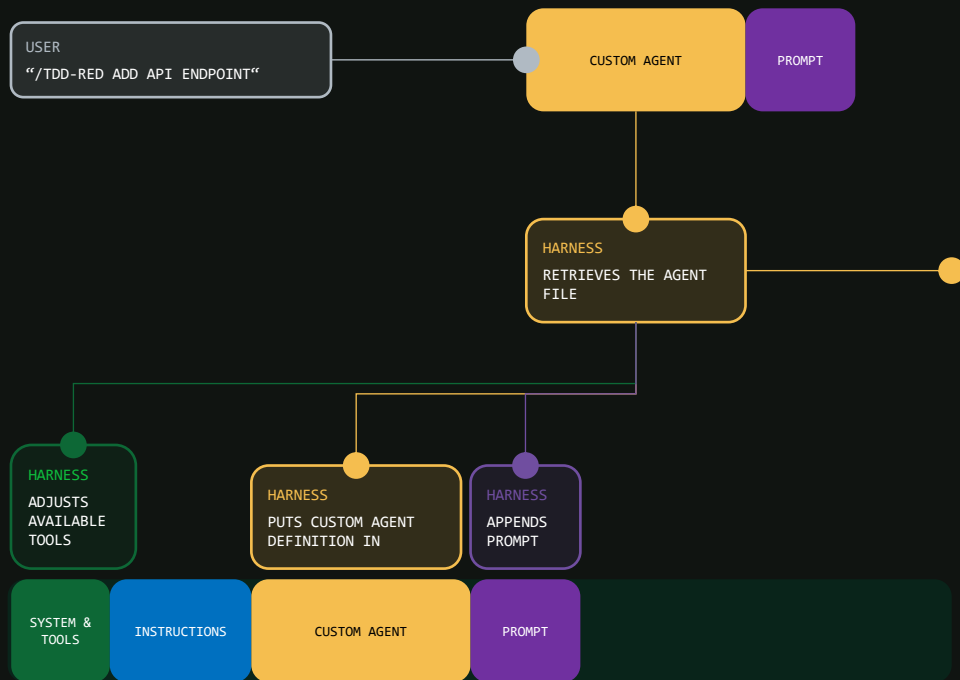
- Keep it very small — re-sent every turn
- Write it yourself, not with AI
- Iterate on real agent behavior
- Recreate it often — it's living



"Be concise" does almost as much as a 50-line skill — and output tokens are the priciest bucket.



Custom Agents



What they are

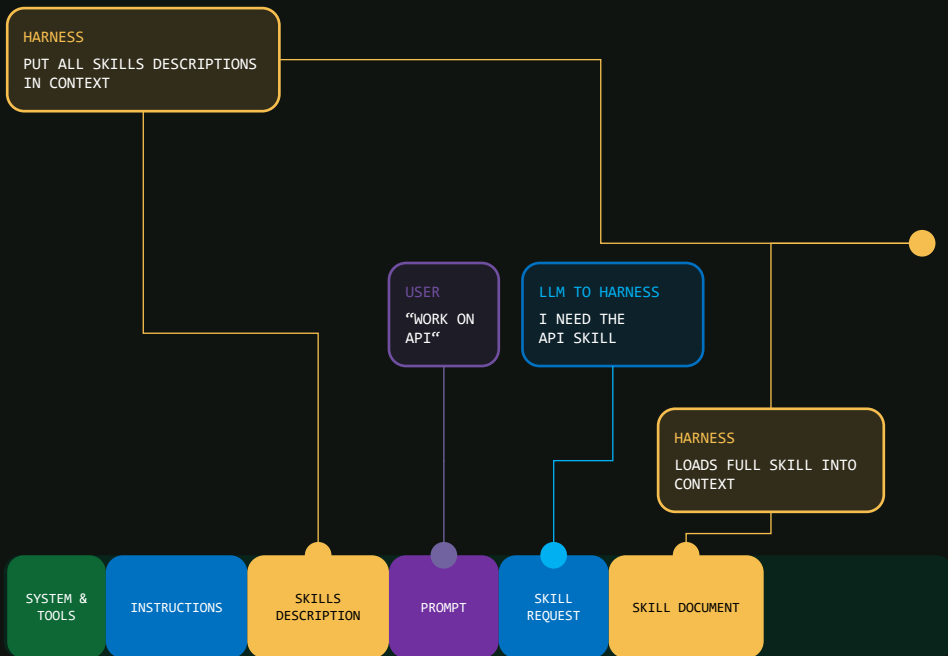
- A reusable prompt that forces a specific way of working
- Manually invoked, e.g. `/tdd-red add API endpoint`
- Write the behavior once; reuse it again and again
- Great for workflows an agent wouldn't choose on its own

Why they help

- The harness adjusts which tools the agent can use
- The real win: preventing wrong paths, not token savings
- Withhold a tool and the agent simply can't misuse it
- e.g. read the issue for a spec — but not edit it

Token savings are minor (tools get cached). The value is control — the agent can't wander somewhere you didn't intend.

Skills



How they work

- Markdown that shapes behavior for a specific task
- Only the short description sits in context up front
- The LLM asks for the skill when the task matches
- The harness then loads the full skill on demand

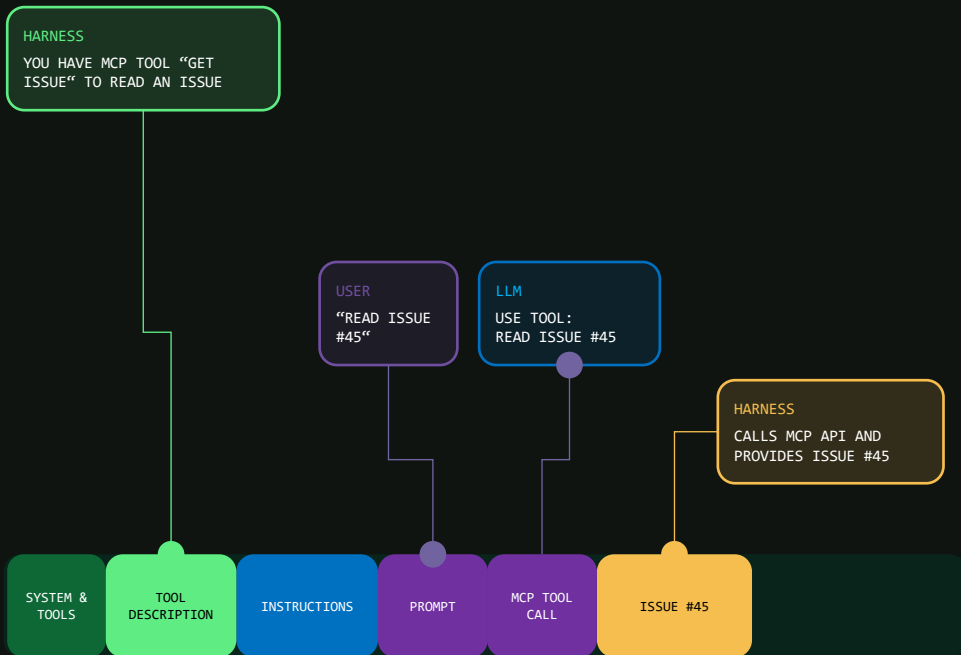
Best practices

- Don't overdo it — you don't need hundreds
- Avoid redundant skills (does it need a "React skill"?)
- Only for capabilities the agent lacks otherwise
- Maintain them — some become unnecessary as models improve

Skills are a great way to offload context that isn't always relevant — pay for it only when the task needs it.

MCP

Integrate with 3rd party tooling



What they add

- External APIs & dynamic tools
- Tool schemas enter the window
- e.g. `get_issue` → "read #45"
- Harness calls API, returns result

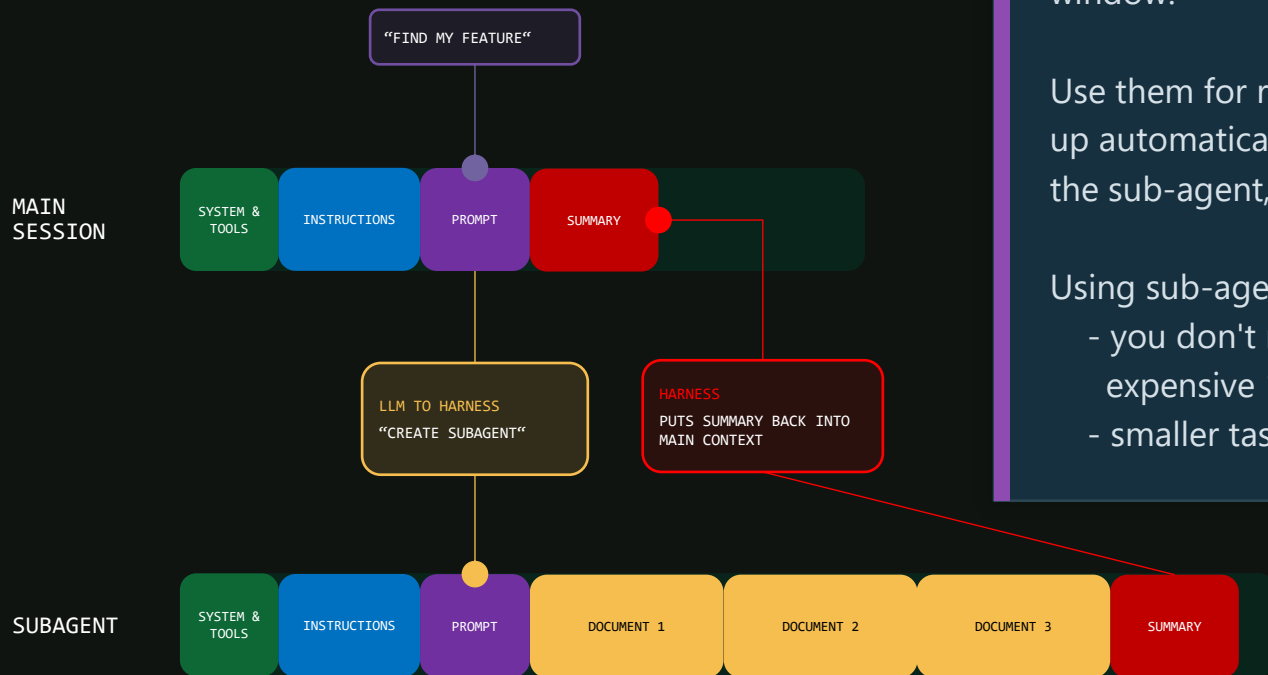
Be rigorous

- They bloat tool schemas every turn
- Can lead agents to wrong tools
- Deactivate ones you rarely need
- Scope heavy ones in custom agents

Playwright is powerful but costly — screenshots and page reads burn tokens. Turn it on only when a task truly needs it.

Subagents

Offload task-specific context



The math that makes sub-agents pay:

a sub-agent reads five files to answer one question, but your main session only gets the final answer — those five files never enter your main window.

Use them for research; the agent often spins one up automatically. Trade-off: tokens are spent in the sub-agent, so it's a conditional optimization.

Using sub-agents will means that:

- you don't need to use the larger and more expensive 1M token context windows
- smaller tasks means less drift – better accuracy

The lighter-weight configs, briefly

Scoped instructions

Conditional instructions based on file-path patterns. Non-deterministic — offered to the agent like a skill. Start static; only scope when instructions get too long.

Prompt files

Manually invoked prompts. Handy to trim the toolset and kick off a custom agent or skill from a common starting point. Not in Copilot CLI — skills/agents are usually better.

Copilot memory

Automatic, small always-on instructions learned from your behavior and team patterns. Works in the background; little to tune, but worth checking periodically.

Lower impact on tokens and quality than the previous configs — but good to know they exist and to review them now and then.

Deterministic controls stop errors from compounding

Without tests — errors stack

- A buggy change lands on a buggy change on a buggy change
- Maybe faster and fewer tokens... at first
- What you actually built is an incident
- Cost: wasted CI/CD, review cycles, human time, a debugging session

With tests — the agent self-corrects

- A failing test says "stop — you can't continue"
- The agent fixes, then builds on a stable base
- It restarts accuracy instead of drifting to 50%
- Linters and security scanners do the same job

Errors compound - they don't average.

A 50-step run that's right only if every step is right is $(\text{accuracy})^{50}$ — at 99%/step $\approx$ 60% success; at 95%/step $\approx$ 8%.
A tiny per-step gain pays off hugely.

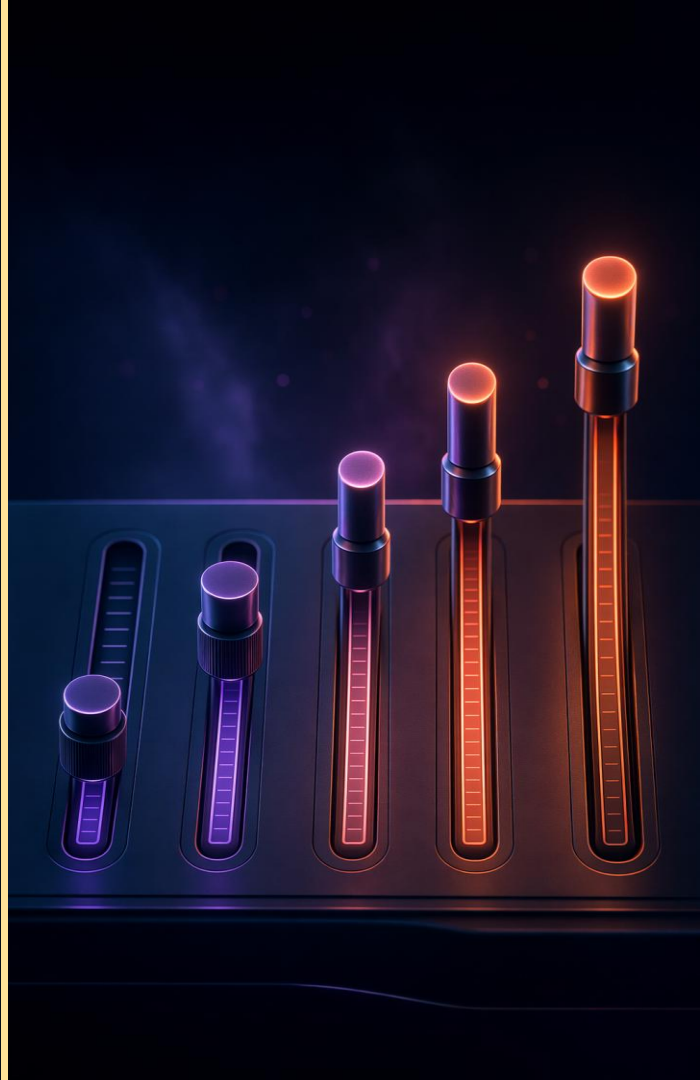
The Copilot CLI team with ~10 devs ships ~500 PRs/week; 53% of their codebase is tests.



SECTION 05

The efficiency playbook

Where tokens get burned, the five levers that stop it, and what the savings look like.



Where tokens get burned — the usual suspects

Reading whole files

"Look at src/" when one function matters — 20k+ tokens.

Long-running sessions

Not clearing between tasks; history rides along.

Oversized instructions

The always-on file, injected every single turn.

Verbose tool output

Piping whole logs in instead of grepping first.

Wrong model for the job

A reasoning model for routine edits.

Re-discovery loops

Re-reading a file after a mid-task /clear.



If you can't explain why Copilot needs a piece of context, it's probably costing you.

Five levers, ordered by leverage

01	Context hygiene	Clear between tasks. Compact proactively. Resume when you can.
02	Prompt discipline	Plan before coding. Reference specific files and lines, not directories.
03	Model selection	Match model tier to task complexity. Choose per phase — a fresh session, not mid-conversation.
04	Scope & tool control	Tight working directory. Content exclusion. Minimal custom instructions.
05	Measurement	/usage, /context, and telemetry make waste visible.

Don't wait for auto-compaction

/clear

Reset the timeline entirely — between unrelated tasks

/compact

Summarize the session, free most of the window — mid-task

/context

Show current token usage broken down by category

/usage

Per-model totals, session duration, files touched

/resume

Reopen a prior session from its saved summary, not zero

/new

Start a fresh session (alternative to /clear)



Sessions are like branches — start a new one per task
and sweep out stale context before it piles up.

Shorter, sharper prompts cut input, output, AND tool calls

EXPENSIVE

```
"Look at my repo and figure out why users get 500 errors on login. Check the auth code, database layer, middleware, and logs. Fix it."
```

Scans many files, reads large chunks, burns tokens before it even narrows down.

EFFICIENT

```
/plan "In @src/auth/login.ts, handleLogin returns 500 when the email has unicode chars. Propose a fix."
```

Targets one file, one function, one case. Plan mode costs one prompt and drives the rest.

Tactics: Plan mode (Shift+Tab) before coding · @file references, not directories · one task per prompt · break big asks into steps

copilot-instructions.md is re-sent every turn — every line is a tax

Keep

- One-line tech-stack context
- Standards the model can't infer from code
- Explicit "do not" rules for files/patterns
- Recurring agent-miss corrections
- A tight file $\approx$ 150 tokens

Cut

- Onboarding essays and team history
- Architecture diagrams as ASCII art
- Full style guides — link instead
- Anything redundant with the code itself
- 3,000 tokens $\times$ every turn $\times$ every dev = real money

LEVER 4 — MATCH THE MODEL TO THE TASK

Reserve premium reasoning for what demands it

Model tier	Best for	Watch for	When to switch
Included / small (GPT-5 mini, 4.1)	Autocompletions, syntax help, boilerplate, simple refactors	Weaker at multi-file reasoning; may hallucinate on big tasks	Default for routine work — often no premium cost
Mid-tier reasoning (Sonnet, GPT-5)	Most everyday coding: features, debugging, code review	The workhorse tier — where most token spend lives	Switch up only when it demonstrably gets stuck
Heavy reasoning (Opus, o-series)	Complex system design, subtle bugs, deep refactors	Steep multiplier; reasoning tokens count as output	Switch down as soon as the hard part is done
Code specialist (Codex)	High-volume code gen; second opinion on tricky diffs	Narrower scope; not the generalist chat model	Pair with a mid-tier for review workflows

Pro move: match the model to each phase — plan → heavy, implement → mid-tier, review → specialist — starting a fresh session per phase. Don't swap models inside a live session: it drops the prompt cache and re-bills the whole context at full price.

LEVER 5 — NARROW THE BLAST RADIUS

Limit what Copilot can see — the most reliable way to stop wasted tokens

/cwd and /add-dir

Explicit working directory shrinks the file surface. Add sibling dirs only when a task genuinely needs them.

Content exclusion

Org/enterprise level for large, generated, or sensitive files: /target, *.log, *.min.js, secrets/. Highest leverage.

.gitignore hygiene

Not a guaranteed filter, but reduces what Copilot indexes — build output, generated code, IDE files.

Tool allow / deny

--allow-tool / --deny-tool and session approvals stop shell commands that would balloon tool output.

Content exclusion is admin-configured and applies to every developer automatically — highest leverage, lowest ongoing cost.

Specialist commands that save tokens

/explore

Fast codebase Q&A. Answers flow back without cluttering the main conversation with file reads.

/task

Runs commands like tests and builds. Reports brief on success, full output only on failure.

/plan

Produces a structured implementation plan before code is written. Cheap, high-impact.

/review

Code review with a tight signal-to-noise profile. Surfaces real issues, skips nits.

/delegate

Hands off to the cloud agent. Long tasks run async and return a PR — not billed against your session.

Why they save: a sub-agent reads five files to answer one question — your main session only gets the answer.

RECOMMENDED ADDITION

Hidden costs in CI — agentic workflows add up quietly

Why CI is different

- Runs on every PR — costs hide in the pipeline
- Run frequency matters as much as per-run cost
- Easier to optimize: fixed in YAML, wins compound

What worked at GitHub

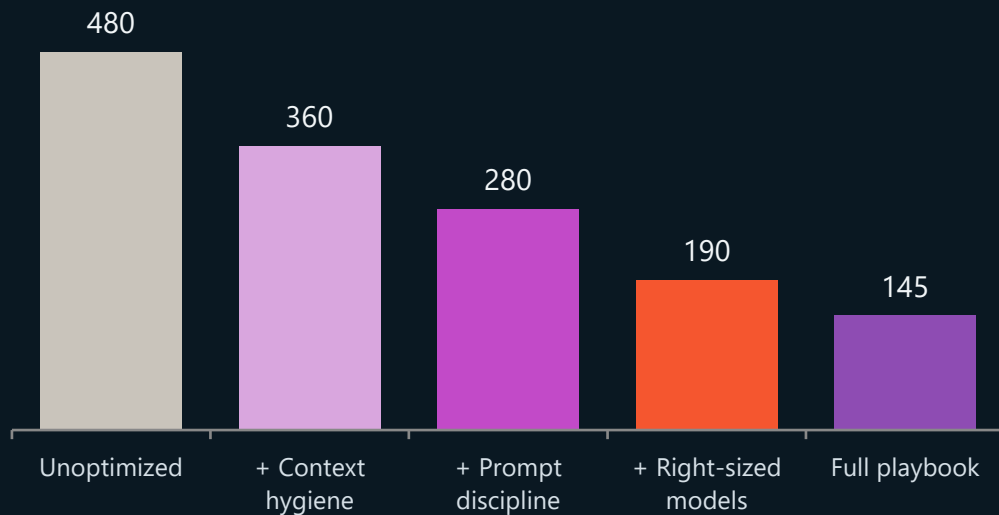
- Daily Auditor flags rising / anomalous usage
- Daily Optimizer files concrete fixes as issues
- Prune MCP tools; move data reads to gh CLI



Biggest win: move deterministic data-gathering out of the LLM loop, into pre-agent CLI steps.

Source: GitHub blog, "[Improving token efficiency in GitHub Agentic Workflows](#)" (May 7, 2026).

What the savings look like when levers stack



The compounding effect

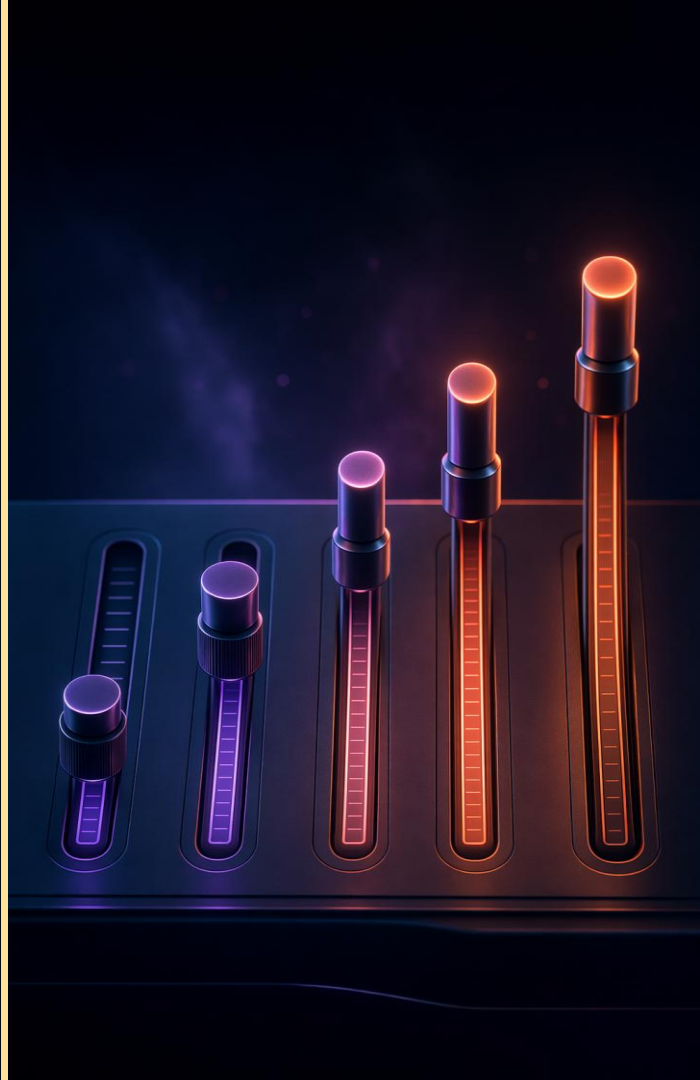
- Each lever removes a different class of waste
- No single tactic is dramatic alone
- Stacked, they routinely cut 60–70% of spend
- Same work ships — you just stop sending junk

Illustrative: a 20-developer team on typical CLI usage. Numbers are not a commitment — run /usage on your own team for a real baseline.



SECTION 06

Three Tools You Can Use Today To Improve Your Usage



Agent Debug Log Summary Page

Agent Debug Logs X

Agent Debug Logs > Add rating feature

Add rating feature

Session Details

Session Type Local
Location Chat Panel
Status Idle
Created 7/3/2026, 9:40:40 AM
Last Activity 7/6/2026, 3:02:37 PM

Summary

Model Turns

82

Tool Calls

108

Total Input Tokens

7,379,130

Total Output Tokens

50,514

Total Cached Input Tokens

6,454,028

Total Tokens

7,429,644

Errors

0

Copilot Usage (AIC)

375.68

Explore Trace Data

View Logs

Agent Flow Chart

Cache Explorer

View Cost Tips in the Copilot CLI using Chronicle

```
> /chronicle
/chronicle standup Get a report on your work from the last day
/chronicle search Search all session content by keyword or topic
/chronicle tips Get personalized tips based on usage patterns
/chronicle cost-tips Get personalized tips to reduce token usage and cost
/chronicle improve Suggest improvements to copilot-instructions.md
```

```
/chronicle [standup|search|tips|cost-tips|improve|reindex]
```

↓ /chronicle cost-tips

07:42

Top cost drivers

1. A few sessions dominate spend.

Evidence: top 3 sessions used 113,175,843 / 187,276,083 tokens (~60.4%); top 5 used ~78.7%; top 10 used ~96.2%. Precision: Exact usage.

2. Very large context windows in long-running chats.

Evidence: largest sessions had massive input vs output (e.g., 54,323,667 in / 254,859 out; 33,392,022 in / 114,291 out), showing repeated re-sending of large context. Precision: Exact usage.

3. VS Code Chat is the primary cost surface.

Evidence: VS Code Chat = 153,989,199 tokens (82.23%); Copilot CLI = 33,286,884 (17.77%). Precision: Exact usage.

4. Oversized pasted content (especially terminal dumps) is recurring.

Evidence: individual user messages up to 60,144 , 45,069 , 44,837 , 40,558 chars, many containing terminal output blocks. Precision: Proxy.

5. High-cost model concentration.

Evidence: gpt-5.3-codex accounted for 83,761,443 tokens (44.73%) overall in scope. Precision: Exact usage.

Ranked recommendations

1. Split long investigations into fresh chats more aggressively

- **Context:** Costs spike when the same conversation carries too much prior context turn after turn.
- **Problem:** Your top sessions were extremely large and context-heavy (top 3 = ~60% of total tokens, with very high input/output ratios).
- **Recommended action:** After each milestone (e.g., diagnosis complete, implementation complete),

C:\lyle\Database\database.cli.llupesms [feature/cli-ratings]

Session: 439 AIC used

Tip: Install the VS Code Extension: “AI Engineer Coach”

The screenshot displays the VS Code interface with the AI Engineer Coach extension installed. The left sidebar shows the extension's menu, with the 'Explore AI Insights' button highlighted by a red box and an arrow. The main panel shows the 'AI Engineer Coach' dashboard, which includes a 'Dashboard' tab, a 'Timeline' view, and a 'Coding Moments' section. The dashboard features a large green circle with the number 82, indicating a score. Below this, there are statistics for 'Ship Goblin Deluxe' (149 REQUESTS, 47 SESSIONS, 35.6K AI LOC, 33 WORKSPACES) and a 'Token Usage & Burndown' section. The 'Anti-Patterns Summary' section shows three metrics: Prompt Quality (77), Session Hygiene (73), and Code Review (90). The 'Tool Mastery' section shows a score of 88. The 'Skill Finder' section is also visible at the bottom.

AI ENGINEER COACH: DASHBOARD

AI Engineer Coach

DETECTED HARNESSSES

- GitHub Copilot App - GitHub
- Copilot CLI - Local Agent

Last synced 7/22/2026, 1:37:09 PM

Explore AI Insights

Sync Sessions

Build 6/29/2026, 2:21:22 PM

AI Engineer Coach X

OBSERVE

- Dashboard
- Timeline
- Coding Moments

MEASURE

- Output
- Patterns

IMPROVE

- Anti-Patterns
- Skill Finder
- Context Health
- Level Up

WORKSPACE

Current

All

Search workspaces...

HARNESS

All Harnesses

82

Ship Goblin Deluxe

You get your own custom title, based on your behaviors!

Dashboard vibes sampled across 4 dimensions

149 REQUESTS

47 SESSIONS

35.6K AI LOC

33 WORKSPACES

Local Agent

GitHub Copilot App

GitHub Copilot CLI

SQL

CSS

CS

MD

YML

C#

PS1

Razor

Token Usage & Burndown temporarily hidden

These features are disabled until we can verify that reported numbers align with GitHub's billing data. They will be re-enabled once validated.

Anti-Patterns Summary

View All Anti-Patterns →

Prompt Quality

77

Good

+9% WoW

MoM 0%

Score improved 9% this week.

Session Hygiene

73

Good

WoW 0%

MoM 0%

Break complex tasks into smaller, focused requests. Use lighter models for simple questions.

Code Review

90

Excellent

WoW 0%

MoM 0%

Set up a .devcontainer/devcontainer.json (or use a Codespace) to sandbox AI-driven terminal execution. Devcontainers isolate builds, installs, and destructive commands from your host OS — especially important when using agent mode with auto-approved terminal access.

Tool Mastery

88

Excellent

+15% WoW

MoM 0%

Stabilize the front of your prompts: keep custom instructions short and stable, avoid frequent compaction, prefer file references over pasted code, and avoid clearing chat mid-task. Within a session, repeated prefixes get cached and re-used for free.

Skill Finder

Scans your prompt history for repeated patterns that waste time re-explaining the same tasks.

Open Full View →

main

Projects: 12

Debug Any CPU

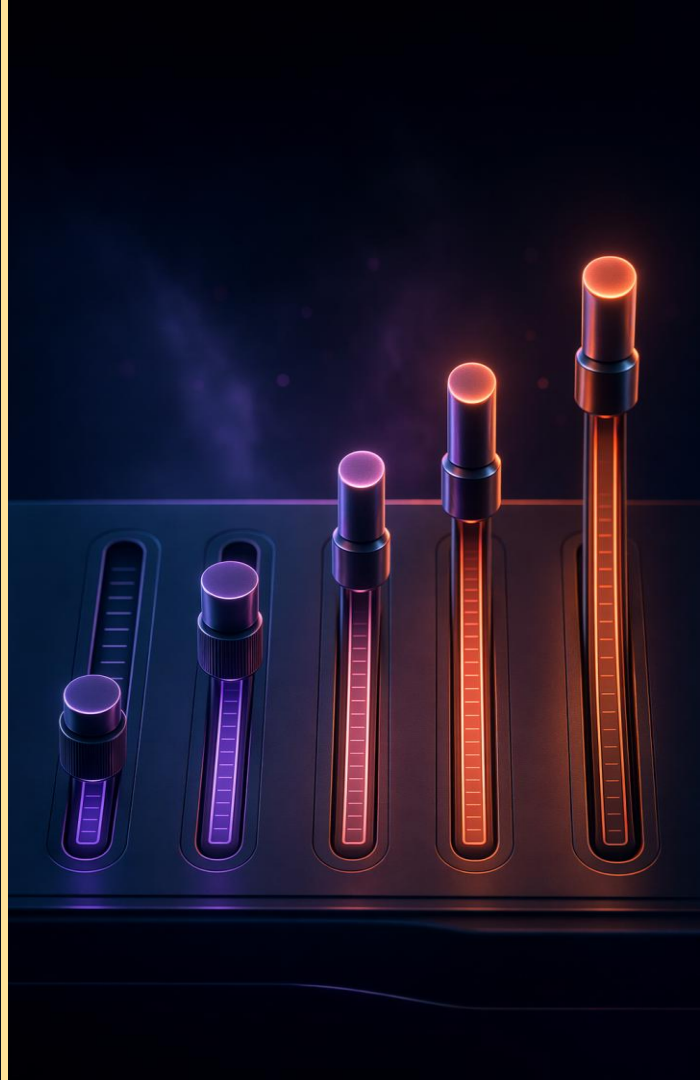
C# Workspace Dev Mode

<https://github.com/microsoft/AI-Engineering-Coach>



SECTION 07

Call to Action



References & further reading

Optimizing GitHub Copilot Cost in the Usage-Based Billing Era

Azure Architecture Blog – July 7, 2026

techcommunity.microsoft.com/blog/azurearchitectureblog

Improve agent quality & token optimization

GitHub Support — product guide

support.github.com/product-guides

Token Economy: Copilot Chat & agents under UBB

Community playbook (aj-enns) · May 2026

github.com/aj-enns/token-economy

Improving token efficiency in Agentic Workflows

GitHub blog · May 7, 2026

github.blog/ai-and-ml/github-copilot

Improving token efficiency for Copilot in VS Code

VS Code engineering blog · June 17, 2026

code.visualstudio.com/blogs

Monitor AI coding agents with Grafana

Microsoft Learn — Managed Grafana + OTel

learn.microsoft.com/azure/managed-grafana

Copilot models, plans & pricing

GitHub Docs — living source of truth

docs.github.com/en/copilot

Tokenomics is the new headcount

Jared Spataro (Microsoft CMO of AI at Work) Blog posted June 2026

www.microsoft.com/en-us/worklab/aiwork-tokenomics-is-the-new-headcount-and-four-more-trends-to-watch

Top ten things you can start doing today

01

Choose the right model for the task:
use Auto mode, escalate only when needed

02

Give clear, focused guidance in your prompts

03

Work in phases - fresh window each time:
Research → Plan → Implement

04

Add deterministic guardrails:
tests, linters, security scans

05

Keep a concise, human-written
copilot-instructions.md

06

Start a new session per task:
/clear or /new between unrelated work

07

Reference files with @path/to/file, not whole
directories

08

End every session with /usage:
know what it cost

09

Open with /plan for anything beyond a
one-line change

10

Start small – start now: run /usage after your next
three sessions. You'll know where to begin.

- *If you remember one line: write as little context as required, and as much as necessary.*



Questions?

Thank you — now go make every token count.

Start small: run `/usage` at the end of your next three sessions. You'll know your baseline and where to begin.

